

Jiayi Xu

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PROFESSIONAL SUMMARY

Software Engineer and Technical Lead with 6 years of experience at Amazon Web Services (AWS) designing, scaling, and operating hyper-scale distributed systems and machine learning infrastructure. Proven track record of architecting cross-organizational ML control planes, modernizing legacy systems, and owning operational excellence from threat modeling to global public release. Passionate about driving autonomous, high-impact technical initiatives.

EXPERIENCE

Amazon Web Services (AWS) – AWS Bedrock-SageMaker Hosting Team

Bellevue, WA

Software Development Engineer

Dec. 2023 – Present

Heterogeneous GPU Endpoint Orchestration

- **Architected a custom weighted routing framework**, integrating customer-defined weight ratios across multiple routing algorithms (e.g., load-aware, random) to ensure precise traffic distribution across diverse hardware capacities.
- **Co-designed and implemented** mixed-GPU control plane workflows, prioritizing strict backward compatibility to enable seamless customer migrations from homogeneous endpoints. This established critical fleet elasticity, mitigating Insufficient Capacity Errors (ICE) during regional hardware shortages.

SageMaker Dynamic LoRA Inference System (Dynamic Rosyfinch)

- **Led a 5-engineer team to architect and launch** the SageMaker Dynamic LoRA Inference system (patent filed), establishing the foundational dynamic loading layer for AWS Bedrock CMI and SageMaker Studio. Orchestrated the system design, delegated technical milestones, and drove the end-to-end release, scaling to support **10k+** adapters per endpoint with **25% better inter-token latency (ITL)** than primary competitors.
- **Engineered a sticky session routing mechanism** that seamlessly handled adapter scaling events and significantly reduced reload latency, driving an **88% reduction in P90 TTFT** (16.9s down to 2.1s) for subsequent requests.
- **Redesigned legacy inference lifecycle components**, eliminating critical race conditions and concurrency bottlenecks that threatened stability at high throughput—all while maintaining backward compatibility.

AWS Bedrock Async API for First and Third-Party Models

- **Architected and implemented** a highly scalable job status orchestrator and internal data APIs for long-running inference tasks using AWS Fargate and AWS Step Functions, actively shaping both low-level and high-level system design documents.
- **Consolidated internal and third-party model workflows** into a unified orchestration layer. Drove cross-organizational consensus on complex interface contracts with external model providers, autonomously stepping in as the primary technical driver during critical absences to maintain the production release schedule.
- **Prevented a production incident** by root-causing and patching a critical idempotency bug, then deployed a non-destructive resource cleanup solution that was subsequently adopted as an integration standard by internal partner teams.

Amazon Web Services (AWS) – AWS Identity-Resource Management Team

Seattle, WA

Software Development Engineer

Apr. 2020 – Dec. 2023

AWS Resource Explorer (Tier 1 Distributed Resource Search)

- **Spearheaded the design and independent implementation** of asynchronous customer onboarding APIs and organizational cell assignment workflows. Managed the global subscription lifecycle, achieving a 99.99% onboarding success rate across 100k+ subscriptions over 6 months, sustained by automated failure-recovery and anti-entropy mechanisms.
- **Developed core components of a cross-region search system** that replicates metadata across regions for low-latency queries. Refactored the end-to-end data pipeline to support a distributed cellular architecture, successfully scaling the system to process over 100k TPS sync requests.
- **Resolved distributed data out-of-order delivery issues** in asynchronous replication workflows by redesigning the data plane's deletion strategy. This architectural shift secured global state consistency and eliminated data inconsistencies in high-concurrency environments.
- **Hardened system availability** by designing and deploying a custom API TPS throttling and monthly quota management framework. This solution successfully protected the underlying Elasticsearch backend from DDoS attacks and abusive usage.

TECHNICAL SKILLS

- **Cloud & Infrastructure:** AWS (SageMaker, Bedrock, ECS, Fargate, EC2, Lambda, Step Functions, API Gateway, S3, CDK)
- **Databases & Search:** DynamoDB, Elasticsearch, OpenSearch
- **ML Inference & Systems:** vLLM, LoRA, Model Inference, Distributed Systems Architecture
- **Languages & Frameworks:** Java, Python, Go, SpringBoot, Smithy

EDUCATION

University of Virginia

Master of Engineering, Computer Engineering

Charlottesville, VA

Sept. 2017 – Dec. 2019

Southeast University

Bachelor of Engineering

Jiangsu, China

Aug. 2013 – Jun. 2017